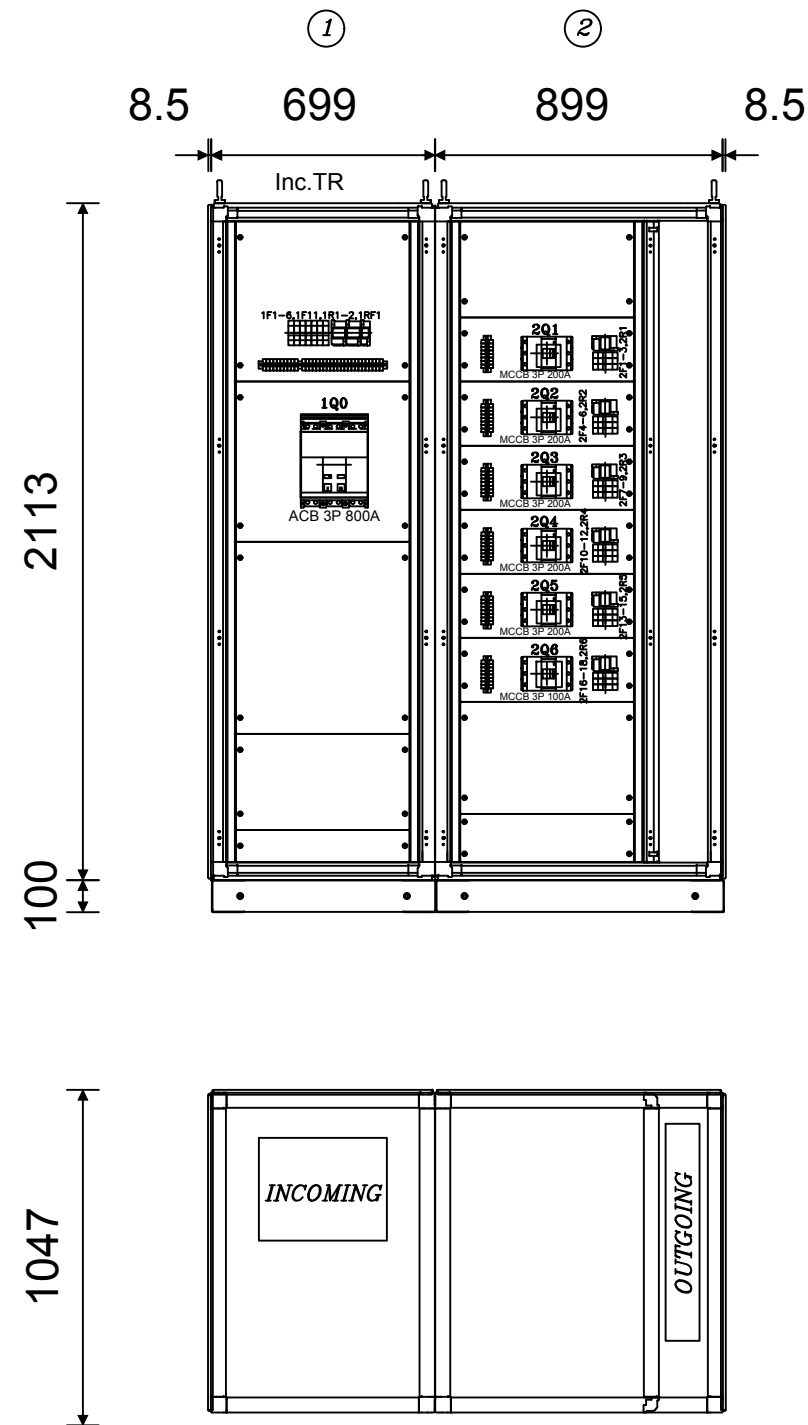
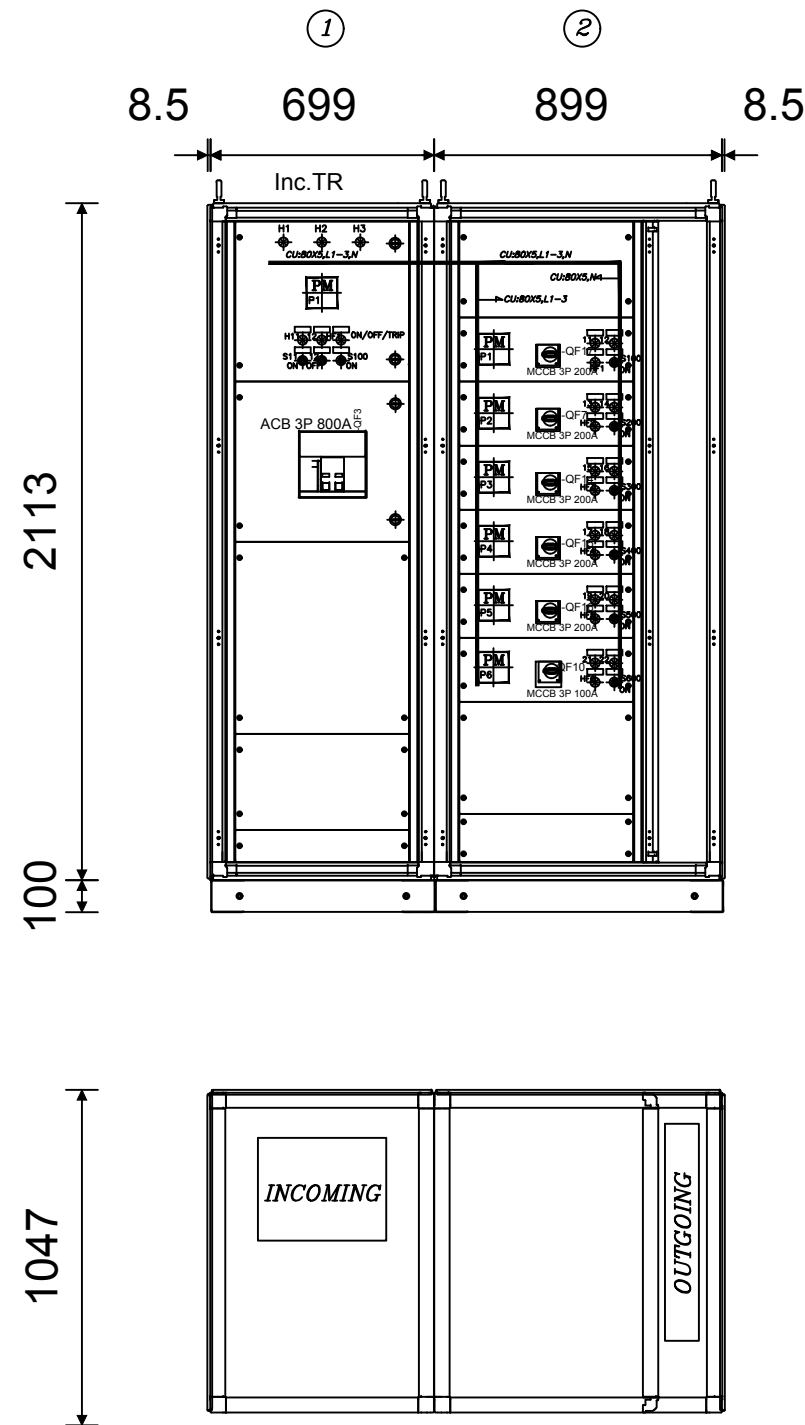


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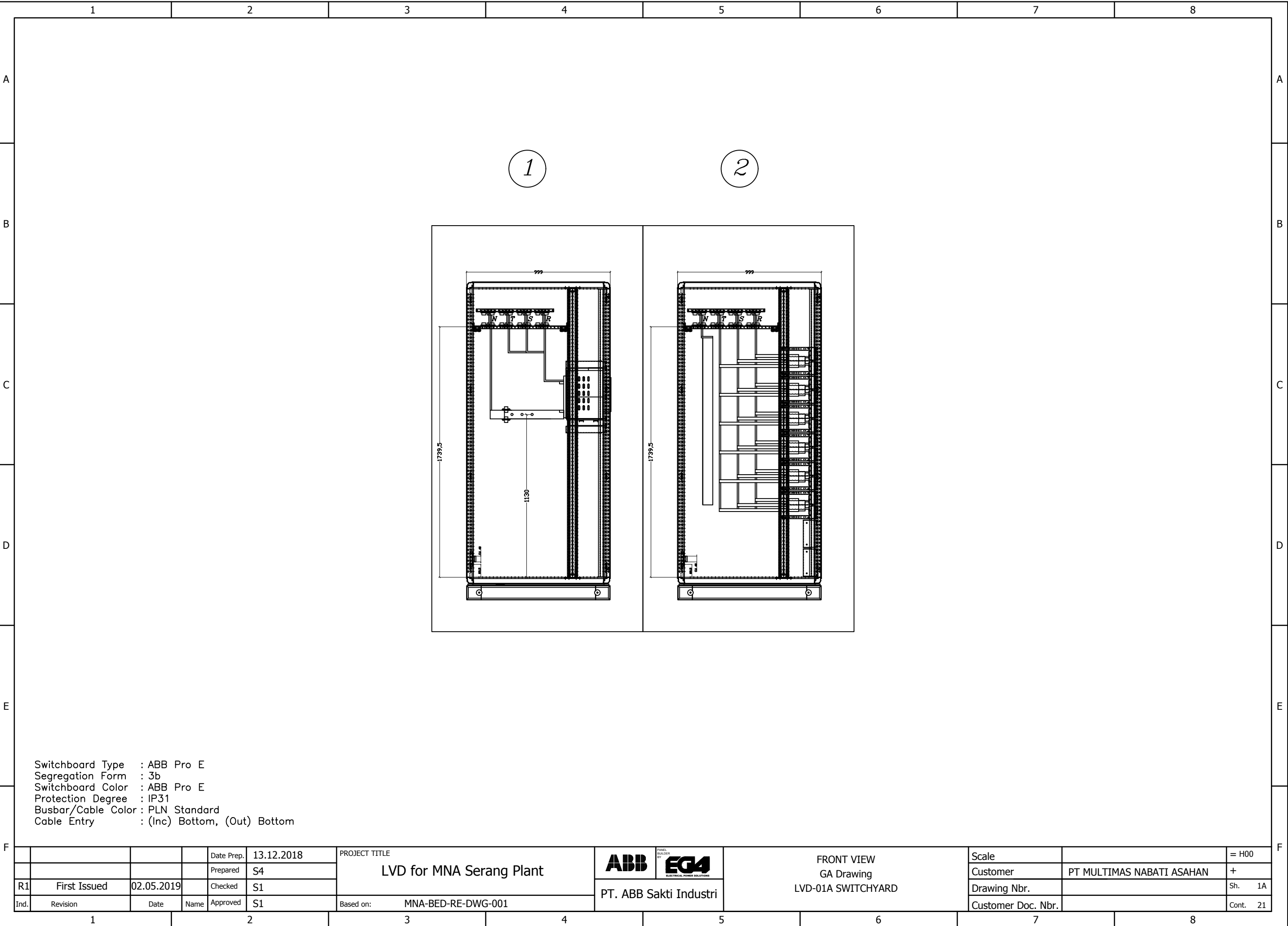
FRONT VIEW
LVD-01A SWITCHYARD



Switchboard Type : ABB Pro E
Segregation Form : 3b
Switchboard Color : ABB Pro E
Protection Degree : IP31
Busbar/Cable Color : PLN Standard
Cable Entry : (Inc) Bottom, (Out) Bottom

				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	  PT. ABB Sakti Industri	FRONT VIEW GA Drawing LVD-01A SWITCHYARD	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

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12345678

A

B

C

D

E

F

SIDE VIEW

BACK VIEW

Switchboard Type : ABB Pro E
Segregation Form : 3b
Switchboard Color : ABB Pro E
Protection Degree : IP31
Busbar/Cable Color : PLN Standard
Cable Entry : (Inc) Bottom, (Out) Bottom

				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDING BY EG4 ELECTRICAL POWER SOLUTIONS PT. ABB Sakti Industri	FRONT VIEW GA Drawing LVD-01A SWITCHYARD	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019		Checked	S1				Drawing Nbr.		Sh. 1B
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.		Cont.	21

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A

B

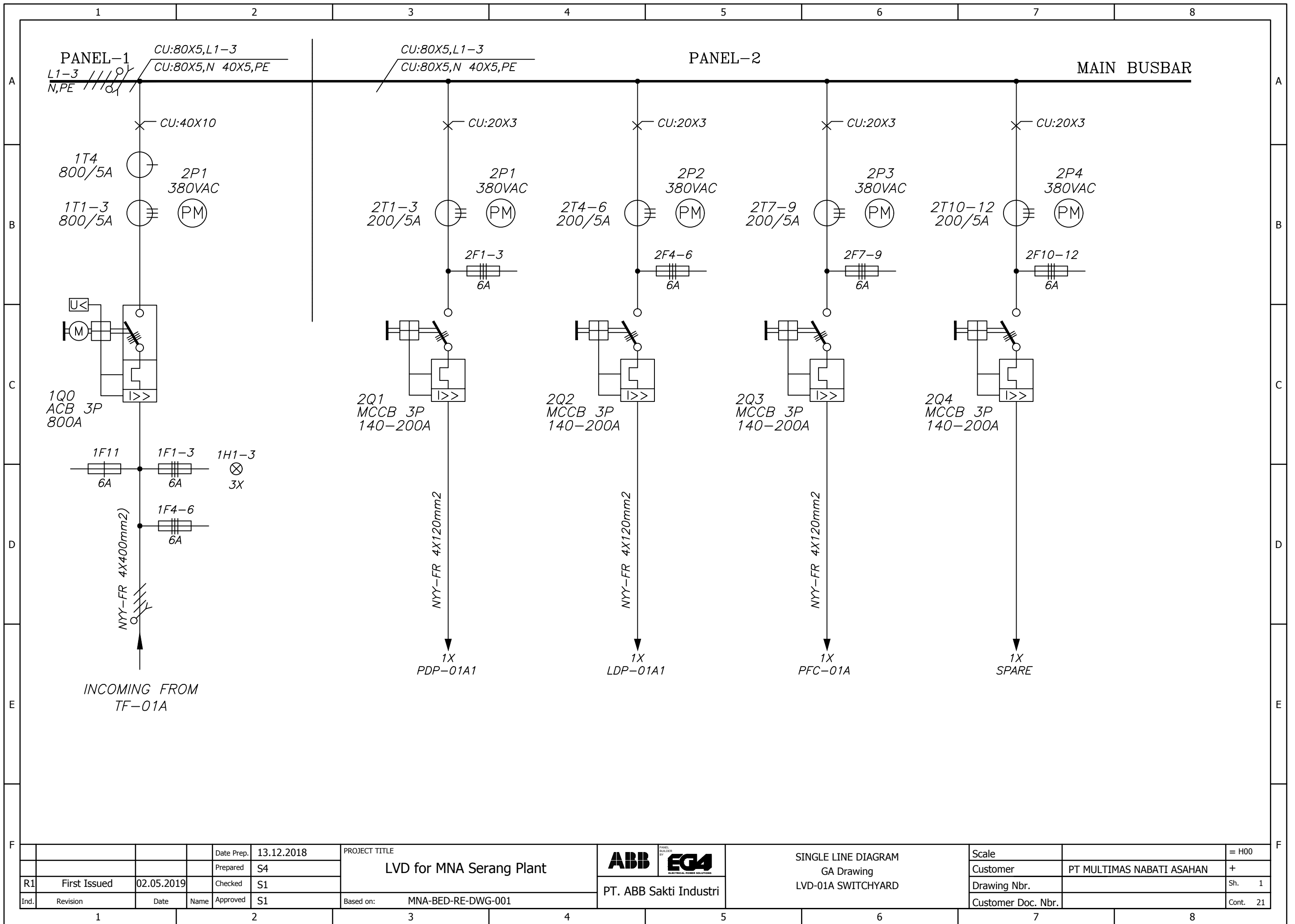
C

D

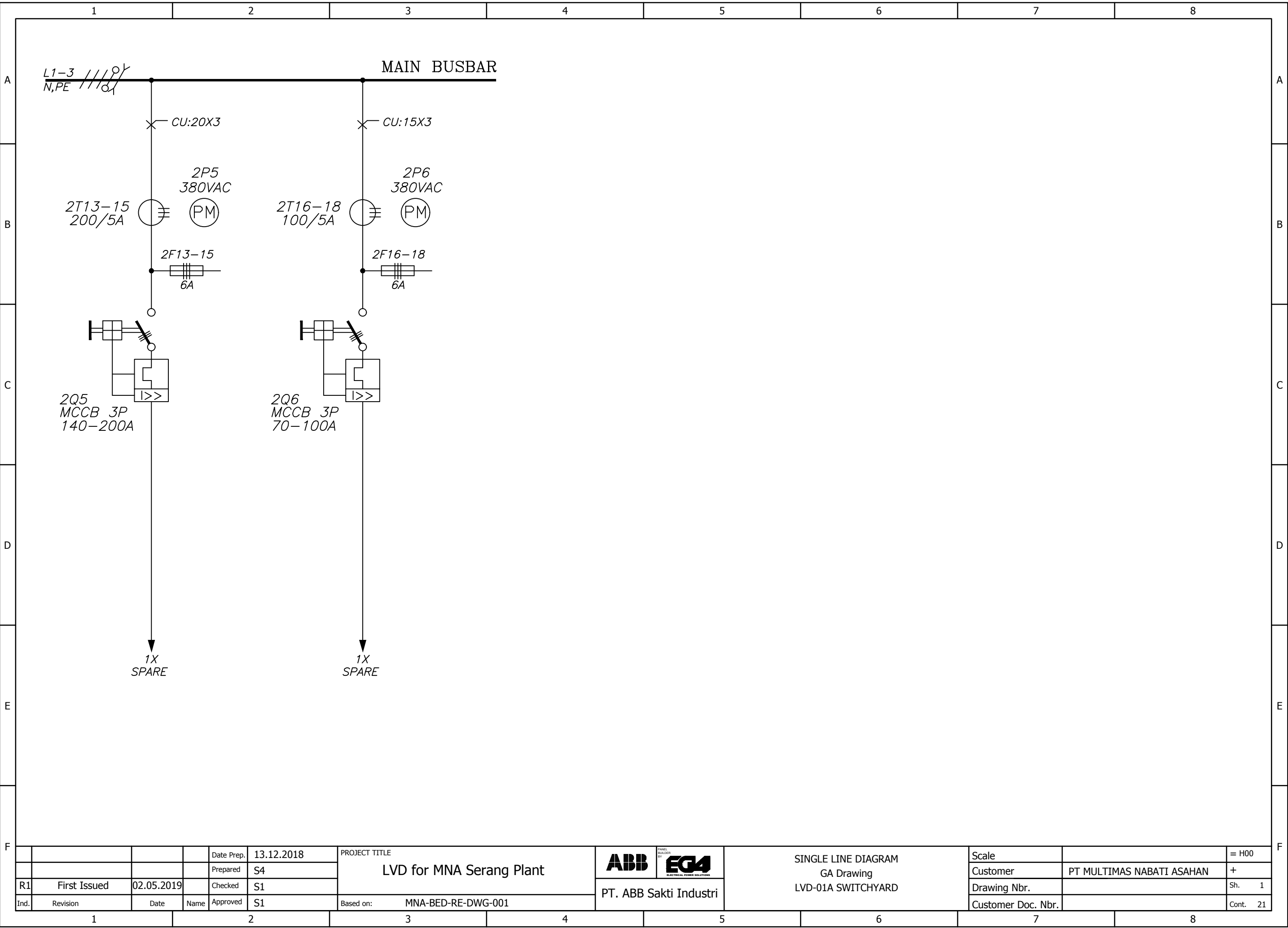
E

F

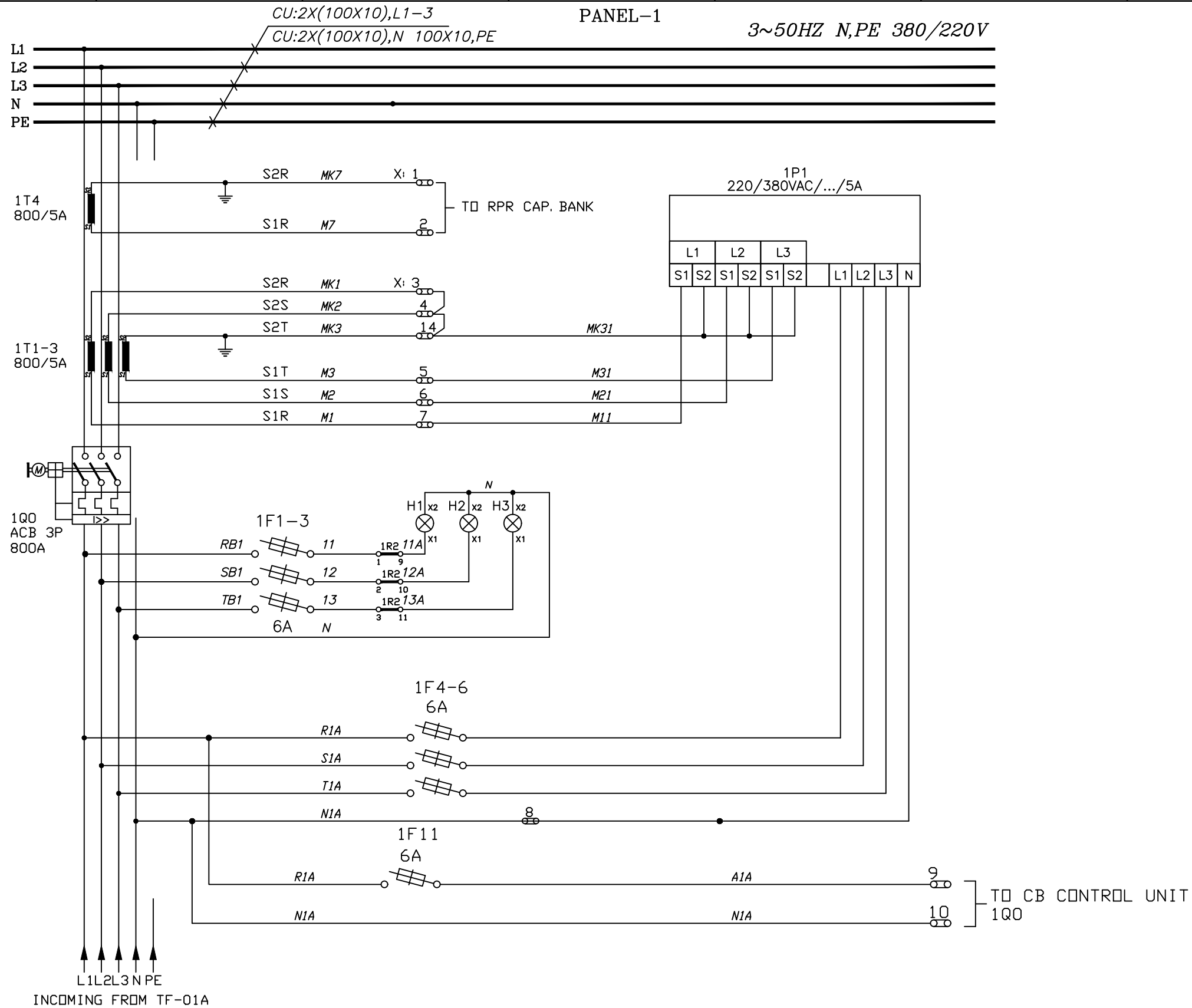
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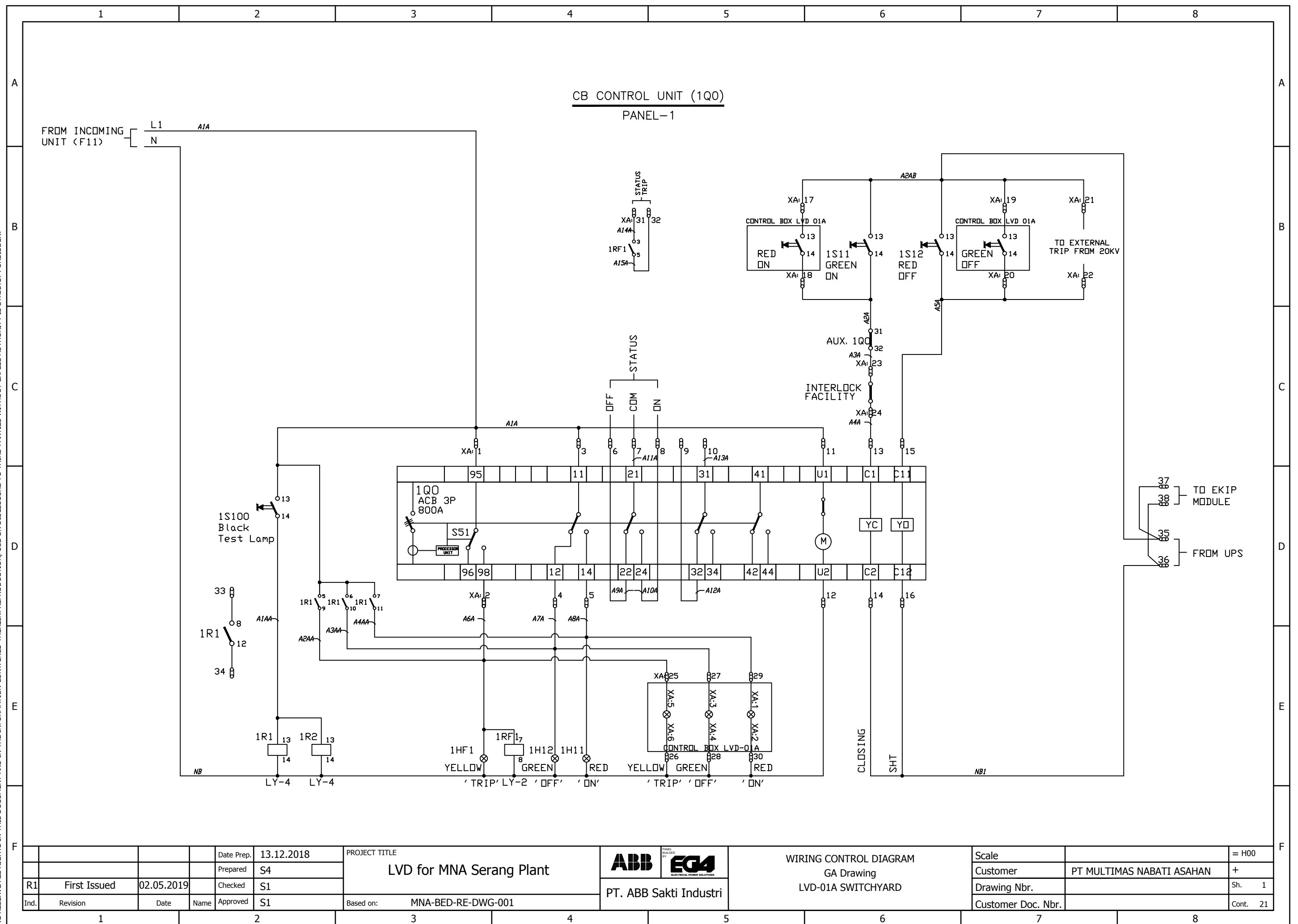
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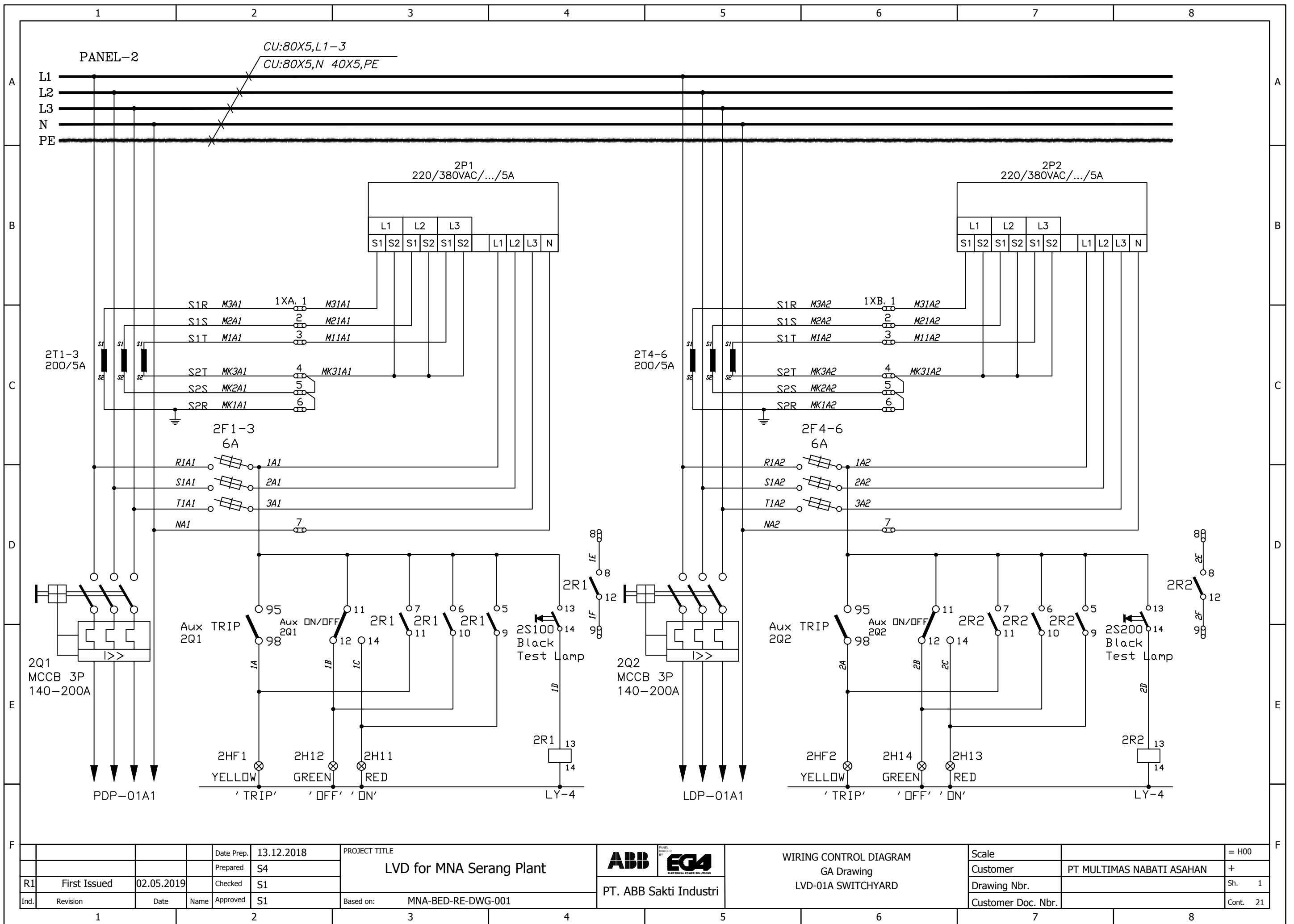
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				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	  PT. ABB Sakti Industri	WIRING CONTROL DIAGRAM GA Drawing LVD-01A SWITCHYARD	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
R1	First Issued	02.05.2019		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

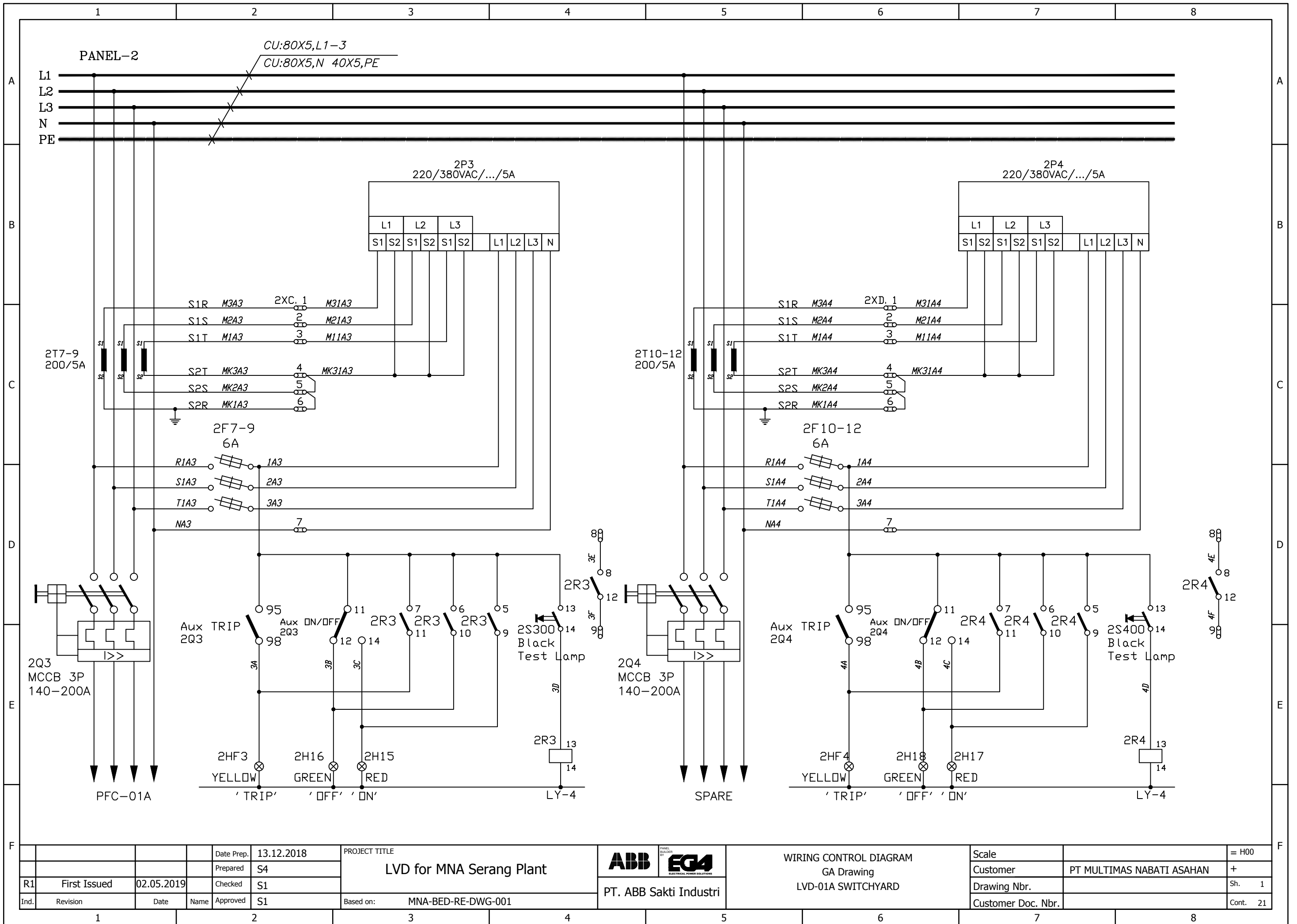


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Date Prep.		13.12.2018		PROJECT TITLE		WIRING CONTROL DIAGRAM		Scale		= H00	
Prepared		S4		LVD for MNA Serang Plant		GA Drawing		Customer		PT MULTIMAS NABATI ASAHAN	
First Issued		02.05.2019		Checked		S1		Drawing Nbr.		Sh. 1	
Ind.		Revision		Date		Name		Approved		Customer Doc. Nbr.	
										Cont. 21	

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			Date Prep.	13.12.2018
			Prepared	S4
R1	First Issued	02.05.2019	Checked	S1
Ind.	Revision	Date	Name	Approved
			S1	

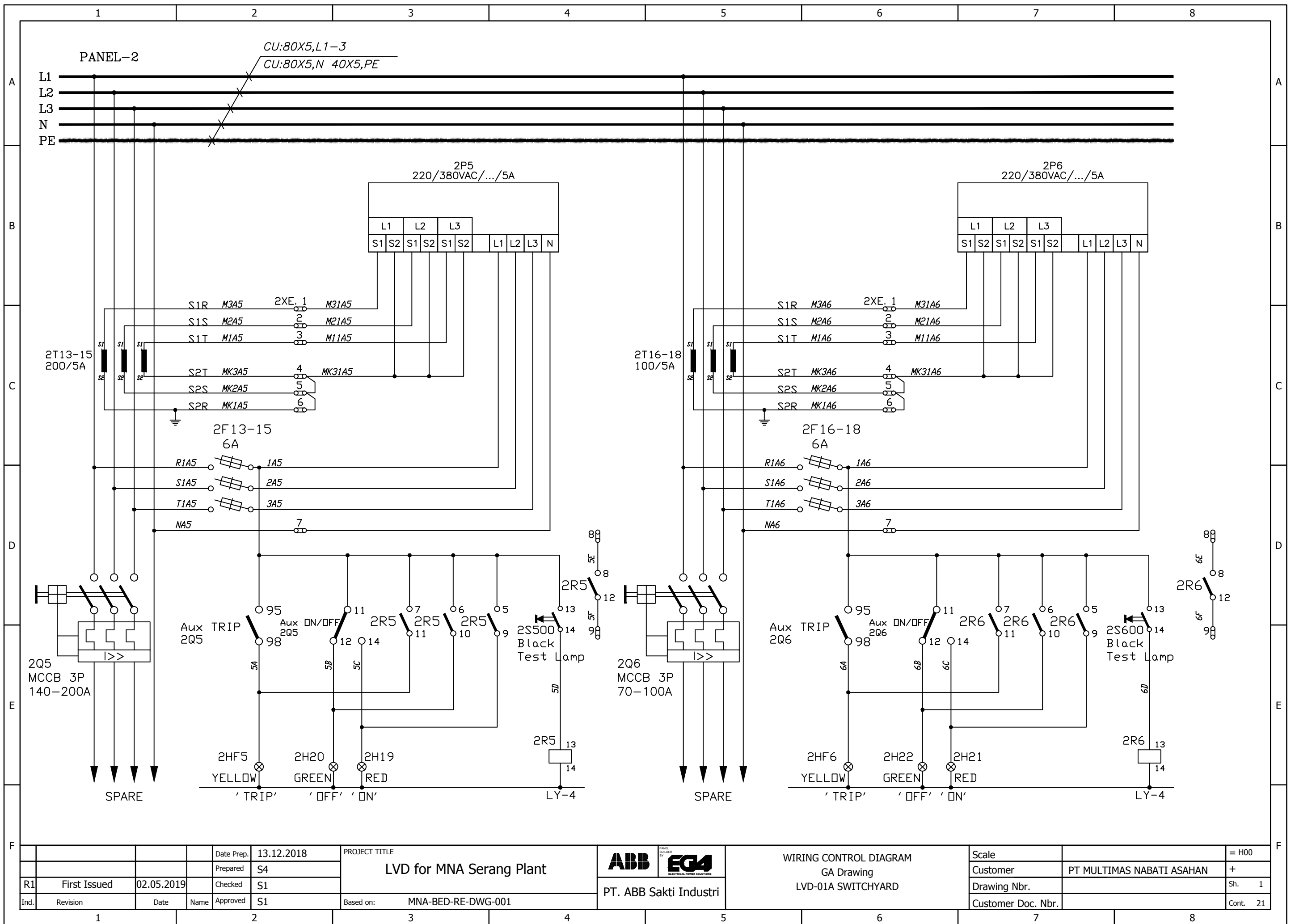
PROJECT TITLE	LVD for MNA Serang Plant
Based on:	MNA-BED-RE-DWG-001

ABB	EG4
PANEL SUBSTITUTION	ELECTRICAL POWER SOLUTIONS
PT. ABB Sakti Industri	

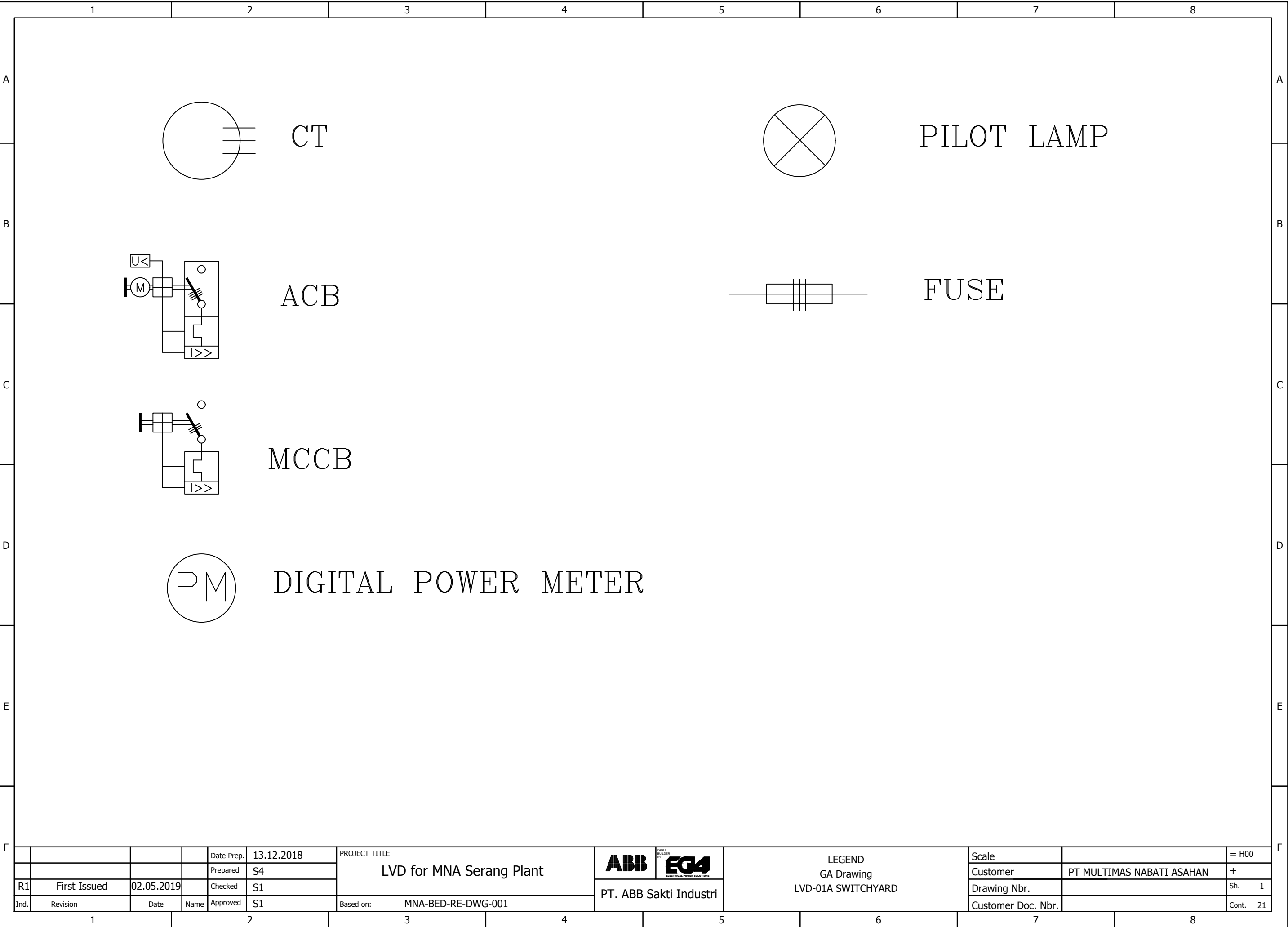
WIRING CONTROL DIAGRAM
GA Drawing
LVD-01A SWITCHYARD

Scale	= H00
Customer	PT MULTIMAS NABATI ASAHAN
Drawing Nbr.	Sh. 1
Customer Doc. Nbr.	Cont. 21

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1		2		3		4		5		6		7		8		
A	N O		DESCRIPTION		TECHNICAL DATA		QTY		DESIGNATION						A	
			INCOMING													
B	1	Fuse Carrier 6A		AXLE/Setara		7		1F11,1F1-7						B		
	2	Pilot Lamp (LED) 220Vac/50Hz, Red		AXLE/Setara		2		1H1,12								
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow		AXLE/Setara		2		1H2,HF1								
	4	Pilot Lamp (LED) 220Vac/50Hz, Green		AXLE/Setara		2		1H3,H11								
	5	DIGITAL POWER METER (*)		ABB, M2M ETHERNET		1		1P1								
	6	CT 800/5A CL.1 10VA		Howig, HG 50I-800		4		1T1-4								
	7	ACB 3P 800A, 50KA, DO, Ekip Touch LSIG (*)		ABB, E1.2C08 D/O		1		1Q0								
C	8	+ Shunt Trip Release (Y0) (*)		ABB, Y0		1		-						C		
	9	+ Motor (M) + Closing Release (YC) For E1.2 (*)		ABB, M+YC		1		-								
	10	+ Padlocks For E1.2 (*)		ABB,		1		-								
	11	+ Ekip Com Protocol IEC 61850 (*)		ABB,		1		-								
	12	Push Button On, 1NO, Green		Ega		1		1S11								
	13	Push Button Off, 1NO, Red		Ega		1		1S12								
	14	Control Relay 220Vac		Omron, LY-2		1		1RF1								
D	15	Control Relay 220Vac		Omron, LY-4		2		1R1-2						D		
	16	Push Button Test Lamp , 1NO, Black		Ega		1		1S100								
	17	OUTGOING														
	18	Fuse Carrier 6A		AXLE		18		2F1-18								
	19	DIGITAL POWER METER (*)		ABB, M2M ETHERNET		6		2P1-6								
	20	Pilot Lamp (LED) 220Vac/50Hz, Red		AXLE/Setara		6		H11,13,15,17,19,21								
	21	Pilot Lamp (LED) 220Vac/50Hz, Yellow		AXLE/Setara		6		HF1-6								
E	22	Pilot Lamp (LED) 220Vac/50Hz, Green		AXLE/Setara		6		H12,14,16,18,20,22						E		
	23	CT 200/5A CL.1 5VA		Howig, HG 30I-200		15		2T1-15								
	24	CT 100/5A CL.1 2.5VA		Howig, HG 30I-100		3		2T16-18								
	25	Push Button Test Lamp , 1NO, Black		Ega		6		2S100-600								
	26	Control Relay 220Vac		Omron, LY-4		6		2R1-6								
															F	
1		2		3		4		5		6		7		8		

				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		ABB PANEL BUILDER BY EG4 ELECTRICAL POWER SOLUTIONS		COMPONENT LIST GA Drawing LVD-01A SWITCHYARD		Scale			= H00	
				Prepared	S4							Customer		PT MULTIMAS NABATI ASAHAN		+
R1	First Issued	02.05.2019		Checked	S1							Drawing Nbr.				Sh.
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001		PT. ABB Sakti Industri		Customer Doc. Nbr.				Cont.	21	

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A <															